U.S. Copyright Office
Software-Enabled Consumer Products Study
Docket No. 2015-6

Comment of Intellectual Property Professors

Introduction

The increasing prevalence of consumer goods that incorporate embedded software highlights two longstanding problems in copyright law—the RAM copy doctrine and the characterization of sales as licenses.[1] Both of these trends arose in cases involving software, but they are now poised to spread beyond that isolated and somewhat idiosyncratic corner of copyright law to the market for consumer goods generally, where they could inflict unforeseen harms on the intellectual property system, the economy, and the public.

Today's consumer products and devices—from smartphones and home appliances to vehicles and medical devices—integrate software code. That code doesn't just offer new bells and whistles that improve on existing products; it is essential to the basic functionality of many devices. Without that software, these products are at best collections of spare parts and at worst expensive paper weights. If consumers are denied their statutory rights to use, transfer, and modify the software at the heart of this new generation of goods, they don't really own the products they buy. Control over embedded software gives copyright holders unanticipated and unprecedented control over how those products are used, if and under what circumstances they can be resold, who if anyone can repair them, and what replacement parts and supplies are compatible with them. While copyright law was intended to give rights holders some measure of control over protected works—subject to principles of exhaustion and fair use, among others—copyright was never meant to provide such a powerful tool to control markets for devices. In this sense, software-embedded goods push copyright law uncomfortably close to regulating utilitarian devices properly within the scope of patent law.

Allowing rights holders to dictate the terms of use, transfer, and repair for tractors, Barbie dolls, and coffee makers also serves to discredit the copyright system in the public mind. Reasonable consumers can be convinced to accept limitations on their ability to download movies without paying for them or photocopy expensive college textbooks. But so far, they appear far less accepting of software that tells them who can repair their tractor or their iPhone or code that restricts which brand of coffee or ink cartridge they can use with their device. On a nearly weekly basis, the popular press reports on a new collision between the well-established expectations of consumers to use the products they buy as they see fit and the altogether new degree of control software code allows rights holders to wield. A copyright law that embraces those restrictions puts its legitimacy at risk.

The challenge that software-embedded goods presents for the copyright system is a direct result of over two decades of software exceptionalism. For a variety of reasons, courts have reshaped copyright doctrine to accommodate the unique nature of software code and the

[1] The anticircumvention provisions of the DMCA present a third, related set of concerns that are the subject of a separate Copyright Office study. See 80 FR 81369. In the context of software-enabled goods, a stark divide between these issues is artificial and obscures the important ways in which they are intertwined. Nonetheless, this comment will make an effort to set aside issues that relate primarily to DRM.

market preferences of the software industry. Unless we are willing to allow that brand of software exceptionalism to infect the digital economy as a whole, it is time to rethink the rules that developed during the infancy of software copyright.

Courts have treated software as a special case for a number of reasons. For one, it has always been an awkward fit within our copyright system. After the legislative acceptance of software copyright in 1980, courts and Congress have felt the need to make a number of accommodations to traditional copyright principles to make room for software. Consider § 102(b) and its exclusion of any "procedure, process, system, [or] method of operation" from the scope of copyright protection.[2] Courts have managed to reconcile protection for software with that fundamental limitation, although their precise relationship remains a contested question.[3] From protectability to infringement, copyright doctrine has been molded and stretched to address the unique problems software presents. Software is different from other works in another respect. Copies of software have little intrinsic value in isolation. Unlike books, paintings, or even musical compositions, the value of software is only realized within an operating environment, where running the code requires loading it into memory. Software also stands apart because the software industry has characterized transactions as licenses since the early days of mass markets for software.[4] Book publishers, record labels, and others attempted to impose licensing restrictions on copies in the past. Consumers and courts saw those efforts as anomalous departures from existing transactional forms and established expectations. Accordingly, they rejected them. But in the software realm, attempts to characterize transactions as licenses have been around from the beginning. A key feature of those early licenses was the idea that the software developer does not sell copies of the program, but merely licenses them.[5] Early efforts to license software reflected the uncertainty surrounding available copyright protection and concerns over confidential trade secret information. But once that uncertainty was resolved, the license remained. It was and remains a powerful tool for undermining exhaustion rules, controlling secondary markets, and limiting competition.

Even if software exceptionalism was justified in the past, it isn't any longer. Digital copies of all sorts share many of the characteristics of software. They are useless outside of a computing environment, and they are subject to pervasive licensing efforts. The distinction between software and other kinds of copyrighted works, which was never particularly clear as a statutory or conceptual matter, is breaking down from a practical perspective.[6] And as computers take the shape of phones, cars, and pacemakers, software is no longer some isolated corner of the marketplace confined to virtual machines. It is a ubiquitous component of the physical world. Equally importantly, the intervening years have provided a clearer picture of the troubling consequences of software exceptionalism.

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[2] 17 USC § 102(b).

[3] Lotus v. Borland, 49 F.3d 807 (1st Cir. 1995); Oracle v. Google, 750 F.3d 1339 (Fed. Cir. 2014).

[4] Watts S. Humphrey, "Software Unbundling: A Personal Perspective," IEEE Annals of the History of Computing 34, no. 1 (2002): 60.

[5] Id.

[6] See A.M. Turing, On Computable Numbers, with an Application to the Entscheidungsproblem, 42 Proc. London Math. Soc. 230 (1936); Allen Newell, The Models Are Broken, The Models Are Broken!, 47 U. PITT. L. REV. 1023, 1033 (1986).

One such consequence is the RAM copy doctrine, which treats temporary instantiations of information in memory that persist for as little as a few seconds as fixed copies for infringement purposes.[7] That misstep has cast the shadow of infringement liability over nearly every use—personal and commercial, private and public—of a digital work. If merely reading a digital book or turning on a device with embedded code is potentially an act of infringement, the economy—from cars and tractors to toys and coffee makers—operates at the mercy of software copyright holders.[8] It means that powering on Mattel's Hello Barbie requires a copyright holder permission, not only for the original purchaser, but for all subsequent users and purchasers as well. And if Mattel decides to clamp down on the secondary market for used toys, it could quite simply refuse to grant permission to aftermarket purchasers to load the software that runs the device. For that matter, so could Ford or Volkswagen.

But the clearest illustration of software exceptionalism is the Ninth Circuit's inconsistent handling of the question of copy ownership in software and non-software cases. On June 7, 2010, the same Ninth Circuit panel heard oral arguments *UMG v. Augusto* and *Vernor v. Autodesk*.[9] Both cases turned on the question of copy ownership; both involved the allegedly infringing resale of lawfully-made copies; both cases featured license terms that imposed similar restrictions and purported to preclude copy ownership by the defendants. But despite those similarities, the panel reached very different conclusions relying on two inconsistent approaches. The best explanation for this disparity is that one case was about software, and the other was not.

In *Augusto*, the court focused on the practical reality of the transaction. But in the software case, *Vernor*, it was concerned almost exclusively with the text announcing restrictions and reservations in the licensing agreement. Had the court applied the *Vernor* standard, *Augusto* would have been decided quite differently: UMG characterized the transaction as a license; it prohibited recipients from transferring the discs to others; and it confined them to "personal" use of the discs. Each of the three *Vernor* factors were satisfied. Instead, the Ninth Circuit created two parallel frameworks for distinguishing licenses from sales. In software cases, it applies a test that turns on factors entirely within the control of the rights holder. In non-software cases, it looks to the nature and consequences of the transaction.

These two mistakes are linked. By finding infringing reproductions around every corner—every time a work is privately displayed, every time a device with embedded code is turned on, every time a copy in long term storage is loaded into short term memory—the RAM copy doctrine necessitates licenses for the most mundane and expected uses of a work. If every act is infringing, every act requires permission. That necessity reinforces the narrative that consumers aren't really buying their software, they are merely paying for permission to use it.

In order to address the pressing issue of software-enabled products, copyright law must undertake a substantial course correction. It must recognize the RAM copy doctrine and the

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[7] See MAI Systems v. Peak Computer, 991 F2d 511 (9th Cir 1993). Courts have started to push back on the most obvious excesses of the RAM copy doctrine. Cartoon Network v. CSC Holdings, 536 F3d 121 (2d Cir 2008). But there is little certainty about its viability and scope. See Aaron Perzanowski, Fixing RAM Copies, 104 Northwestern L. Rev. 1067 (2010).

[8] See Jessica Litman, The Exclusive Right to Read, 13 Cardozo Arts & Ent. L.J. 29, 31-32 (1994).

[9] UMG Recordings, Inc. v. Augusto, 628 F.3d 1175 (9th Cir. 2011); Vernor v. Autodesk, Inc., 621 F. 3d 1102 (9th Cir. 2010).

widespread acceptance of the consumer license as costly mistakes. And it must reaffirm the strong commitment to balancing the exclusive rights of copyright holders with the personal property interests of consumers reflected throughout the Copyright Act. If instead, copyright law continues down the existing path of software copyright, those mistakes will be amplified throughout the economy.

1. The provisions of the copyright law that are implicated by the ubiquity of copyrighted software in everyday products

At least six provisions of the Copyright Act are implicated by software-enabled devices.

Section 101 provides that "[a] work is 'fixed' in a tangible medium of expression when its embodiment in a copy or phonorecord, by or under the authority of the author, is sufficiently permanent or stable to permit it to be perceived, reproduced, or otherwise communicated *for a period of more than transitory duration*."[10] (emphasis added). By treating the very act of loading software into memory when a device or product is powered on or used, the RAM copy doctrine ignores this durational requirement and introduces unnecessary uncertainty about the legality of consumers using the devices they own.

Section 102(b) excludes "any idea, procedure, process, system, method of operation, concept, principle, or discovery" from the scope of copyright protection.[11] Among the many goals of § 102(b) is maintaining the boundary between the copyright and patent systems.[12] A device maker should not, by virtue of their copyright interest in a software program, be permitted to control the use, transfer, and repair of a device. As the Court explained in *Baker v. Selden*, "That is the province of letters patent, not of copyright."[13] But without meaningful exhaustion and corresponding consumer property rights, that is precisely the control copyright offers for makers of software-enabled devices.

Section 106(3) defines the types of transactions available to copyright holders under the exclusive right of public distribution. It provides that copyright holders have the exclusive right "to distribute copies or phonorecords of the copyrighted work to the public by sale or other transfer of ownership, or by rental, lease, or lending."[14] This language recognizes two categories of transactions: (1) sales and other transfers of ownership, like gifts; and (2) transfers of limited duration: rental, lease, or lending. Some transactions commonly referred to as licenses are easy to fit into category (2). Borrowing a library book or subscribing to Netflix doesn't transfer ownership. Other transactions, like the exchange of perpetual possession of a physical copy in exchange for a one-time payment, fit more naturally into category (1), regardless of how they are characterized in license terms. In any case, the text of the Act makes clear that there is no free-standing transactional form called a "license" when it comes to the transfer of particular copies, such as those embedded in a phone, watch, or tractor.

[10] 17 USC § 101.

[11] 17 USC § 102(b).

[12] See Pamela Samuelson, Why Copyright Law Excludes Systems and Processes From the Scope of Its Protection, 85 Tex. L. Rev. 1921 (2006).

[13] Baker v. Selden 101 U.S. 99 (1879).

[14] 17 USC § 106(3).

Section 202 reinforces the distinction between "[o]wnership of a copyright," which the rights holder retains even after the sale of an embedded copy, and "ownership of any material object in which the work is embodied."[15] Nonetheless, license agreements and the courts that enforce them often lose sight of this basic distinction. A rights holder can license certain uses of their copyrighted work at the very same time they transfer ownership through the sale of a particular copy stored in a device or product.

Section 109(a) reflects the longstanding first sale rule, limiting the power of copyright holders to interfere with downstream transfers of copies owned by consumers, and § 109(b) ensures that the owner of a copy is entitled to display it publicly.

Section 117 secures additional rights for owners of copies of computer programs. It permits the creation of copies essential to the use of the program, archival copies, and adaptation copies necessary to translate between programming languages, move the program from one device to another, and to add new features.[16] The statute also permits the transfer of exact essential step and archival copies as part of a transfer of the original copy, triggering a duty to delete any remaining archival copies.[17]

Together, §§ 109 and 117 define the rights of owners of copies.[18] Applying the principle of copyright exhaustion, these two provisions demonstrate the steps the Copyright Act takes to preserve the personal property interests of owners. Section 117, like § 109, reflects congressional recognition of the fact that copyright has the potential to interfere with the legitimate personal property interests of owners of copies. That potential conflict is even greater when software leaves the familiar confines of our computers and makes its way into nearly every electronic device that surrounds us.

To solve that problem, § 117 recognizes a set of rights that belong to copy owners. But that solution only works in a legal environment that acknowledges the purchasers of software products and software-embedded devices as owners. Too many courts have ignored this congressional purpose and fallen for the software industry's myth of licensed copies. In doing so, those courts have created a legal framework that bestows on copyright holders set of rights the Copyright Act was simply never intended to grant.

2. Whether, and to what extent, the design, distribution, and legitimate uses of products are being enabled and/or frustrated by the application of existing copyright law to software in everyday products

A range of legitimate uses are frustrated by the application of the RAM copy doctrine and the acceptance of software transactions as licenses rather than sales. Combined, those doctrines give rights holders the power to control whether and how consumers use the devices they own; when and under what conditions they can lend, resell, or give them away; who can repair

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[15] 17 USC § 202.

[16] 17 USC § 117(a); National Commission on New Technological Uses of Copyrighted Works, Final Report 13 (1979).

[17] 17 USC § 117 (a)-(b).

[18] 17 USC §§ 109 & 117.

them;[19] what interoperable products, replacement parts, and components can be used with them; the degree to which those products can be researched and tested; the possibility of tinkering and user innovation; and the availability of interoperable products.

Below are just a few examples of how control over the software embedded in consumer devices frustrates their legitimate use. Many of the harms these examples represent are compounded by the anticircumvention provisions of the DMCA.[20] But even putting § 1201 aside, the unauthorized use, transfer, and modification of copies of software arguably violate § 106 if we accept the RAM copy doctrine and reject consumer ownership of embedded software copies.

- Amazon used the software embedded in its Kindle devices to remotely delete purchased books—including George Orwell's 1984—and to disable consumer accounts, effectively rendering the devices unusable.[21]
- Nintendo used the software embedded in its Wii U gaming device to force owners to accept a revised end user license agreement. Device owners were given no option to refuse the update. The software prevented all attempts to access games or other data until device owners clicked "Agree" to the new terms.[22]
- Google's terms of use for its Glass wearable computer prohibit owners from reselling, loaning, transferring, or giving away the device. And its embedded software allowed Google to remotely deactivate the device in the event of an unauthorized transfer.[23]
- Cisco insists that owners of its hardware are not entitled to transfer copies of the software they contain, software that is essential to their function. Device owners, Cisco claims, "[do] not own the software on the product - and therefore [have] no rights to sell

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[19] Section 117 permits the owner or lessee of a machine to create a copy of software for the purposes of repair. 17 § USC 117(c). However, the provision contains two crucial limitations that severely narrow its scope and has never been applied to common consumer goods.

[20] Some have been addressed by exemptions granted in the recent § 1201 rulemaking, but such exemptions do not preclude infringement liability, nor do they permit modification of the underlying code. Moreover, those exemptions are temporary, requiring renewal every three years, and the exemption related to motor vehicles does not go into effect until October 2016.

[21] Brad Stone, Amazon Erases Orwell Books from Kindle, New York Times, July 17, 2009, http://www.nytimes.com/2009/07/18/technology/companies/18amazon.html; Mark King, Amazon Wipes Customer's Kindle and Deletes Account with No Explanation, Guardian (UK), October 22, 2012, http://www.theguardian.com/money/2012/oct/22/amazon-wipes-customers-kindle-deletes-account.

[22] Kit Walsh, Nintendo Updates Take Wii U Hostage Until You "Agree" to New Legal Terms, Deeplinks, Oct. 14, 2014, https://www.eff.org/deeplinks/2014/10/nintendo-updates-take-wii-u-hostage-until-you-agree-new-legal-terms.

[23] David Kravets & Roberto Baldwin, Google Is Forbidding Users from Reselling, Loaning Glass Eyewear, Wired, April 17, 2013, http://www.wired.com/2013/04/google-glass-resales; Long Comment Regarding a Proposed Exemption Under 17 U.S.C. 1201 (Proposed Class # 21) at 6, Section 1201 Exemptions to Prohibition Against Circumvention of Technological Measures Protecting Copyrighted Works: Second Round of Comments, accessible at http://copyright.gov/1201/2015/comments-032715/class%2021/John_Deere_Class21_1201_2014.pdf.

it to you. This means that if you purchased the product from that seller, your company ... would be in violation of Cisco's intellectual property rights."[24]

- John Deere tractors contain a variety of control units—hardware and software components that regulate functions ranging from running the engine to adjusting the armrest. John Deere leverages its control over that software to prevent home repairs and shut out independent repair shops. Instead, tractor owners are forced to pay for expensive service at authorized John Deere dealers. John Deere defends this policy on the grounds that its customers don't actually own the software embedded in their tractors. In John Deere's estimation, tractor owners merely have "an implied license for the life of the vehicle to operate the vehicle."[25]
- Apple created software for the iPhone that detects "unauthorized" repairs of the devices. For iPhone owners who choose to opt for an independent repair shop, the consequences can be catastrophic. The device displays the cryptic "Error 53" and renders the device unusable and the data stored on it inaccessible. To date, Apple has announced no remedy to fix Error 53. As a result unknown numbers of iPhone owners have seen their devices rendered worthless by Apple's software.[26]
- Companies like Keurig have relied on embedded software to control the market for consumable products like coffee pods. The DMCA reinforces these software restrictions, but § 1201 is not necessary to restrain consumer choice. A software license that prohibits modification or removal of the lockout software could trigger traditional infringement liability under the RAM copy doctrine.[27]
- Similarly, companies like Apple have relied on embedded software to prevent device owners from installing or using interoperable products. Only approved applications from Apple's app store can be installed on the iPhone and iPad, for example. Modifying a device's operating system to avoid these restrictions violates license restrictions and could constitute infringement.[28] Concerns about potential infringement liability can also discourage research and testing of consumer devices. As recent safety recalls of Ford and Chrysler vehicles demonstrate—along with Volkswagen's environmentally disastrous defeat device—the software that operates our nation's vehicles requires outside scrutiny. But carmakers insist that car owners have no right to inspect or modify that software. GM, for example, recently told the Copyright Office that car owners mistakenly "conflate ownership of a vehicle with ownership of the underlying computer software in a vehicle." In a world that embraces the RAM copy doctrine and refuses to

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[24] Cisco, Third Party Maintenance Services, http://www.cisco.com/c/dam/en_us/about/doing_business/legal/service_descriptions/docs/Third_Party_Maintenance_Services_FAQ.pdf

[25] Kyle Wiens, We Can't Let John Deere Destroy the Idea of Ownership, Wired, April 21, 2015, http://www.wired.com/2015/04/dmca-ownership-john-deere.

[26] Cory Doctorow, Error 53: Apple remotely bricks phones to punish customers for getting independent repairs, BoingBoing, Feb. 5, 2016, http://boingboing.net/2016/02/05/gerror-53-apple-remotely-bric.html.

[27] Hayley Peterson, Keurig's New Coffee Machine Got A Ton Of Hype, But Customers Seem To Hate It, Business Insider, Nov. 25, 2014, http://www.businessinsider.com/keurig-20-scathing-reviews-2014-11.

[28] Tom Krazit, Apple: iPhone Jailbreaking Violates Our Copyright, Cnet, Feb. 13, 2009, http://www.cnet.com/news/apple-iphone-jailbreaking-violates-our-copyright.

recognize consumer ownership of software copies, the specter of copyright liability stands in the way of this important work.[29]

- Current software copyright law can also interfere with user innovation. Owners of medical devices like pacemakers and insulin pumps are understandably motivated to test and improve the devices that are implanted in their bodies and keeping them alive. But because they are not considered the owners of the embedded software copies that make these modern devices work, they are denied the opportunity to innovate even when their lives rest in the balance. In some instances, embedded software even denies patients access to data about their own health.[30]

These examples make clear that copyright law frustrates a host of socially valuable, legitimate uses. Rejecting an approach that treats all RAM instantiations as copies and recognizing owners of devices as owners of the copies of software embedded in them would go a long way towards addressing these and similar scenarios.

With regard to interoperable products, the Federal Circuit's recent opinion in *Oracle v. Google* introduces a related but distinct set of concerns.[31] In finding the structure, sequence, and organization of Oracle's Java API copyrightable, that opinion significantly increases the barriers facing developers of products that interoperate with existing devices and software platforms. First, it misinterpreted § 102(b) as nothing more than a restatement of the idea/expression distinction, giving no independent meaning to the statutory exclusion of procedures, processes, systems, and methods of operation. In doing so, the Federal Circuit misapplied Ninth Circuit precedent by limiting consideration of interoperability to the fair use inquiry. But in *Sega v. Accolade*, the court based its holding on the fundamental premise that software elements necessary to achieve interoperability are beyond the scope of copyright protection.[32] Second, the Federal Circuit adopted an unreasonably narrow interpretation of the merger doctrine when it reasoned that merger was inapplicable so long as Oracle's own engineers could exercise creative judgment in the creation of the API; the choices left available to developers of follow-on interoperable software or devices was, in the Federal Circuit's estimation, irrelevant.[33] This

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[29] Andy Greenberg, After Jeep Hack, Chrysler Recalls 1.4M Vehicles for Bug Fix, Wired, July 24, 2015, http://www.wired.com/2015/07/jeep-hack-chrysler-recalls-1-4m-vehicles-bug-fix; Alisa Priddle, Ford Recalls 433,000 Cars Because Engines Won't Shut Off, USA Today, July 2, 2015, http://www.usatoday.com/story/money/cars/2015/07/02/ford-recall/29620683; Andrea Peterson and Brian FungThe Tech Behind How Volkswagen Tricked Emissions Tests, Washington Post, Sept. 22, 2015, https://www.washingtonpost.com/news/the-switch/wp/2015/09/22/the-tech-behind-how-volkswagen-tricked-emissions-tests; Comments of General Motors LLC (Proposed Class #21) at 10, Exemption to Prohibition on Circumvention of Copyright Protection Systems for Access Control Technologies, No. 2014-07 (U.S. Copyright Office March 27, 2015), accessible at http://copyright.gov/1201/2015/comments-032715/class%2021/General_Motors_Class21_1201_2014.pdf.

[30] Emily Singer, Getting Health Data from Inside Your Body, MIT Technology Review, November 22, 2011, http://www.technologyreview.com/news/426171/getting-health-data-from-inside-your-body; Andrew Gregory and Graham Morrison, Karen Sandler: Full Interview, TuxRadar, September 26, 2013, http://www.tuxradar.com/content/karen-sandler-full-interview.

[31] Oracle v. Google, 750 F.3d 1339 (Fed. Cir. 2014)

[32] See Sega Enters., Ltd. v. Accolade, Inc., 977 F.2d 1510, 1522 (9th Cir. 1992).

[33] Oracle, 750 F.3d at 1361.

result is inconsistent with the prevailing view in many circuits, including the Ninth.[34] The Federal Circuit's rewriting of software copyright law gives copyright holders far greater power to limit interoperability not only with software code, but with the consumer devices that depend on that code.

3. Whether, and to what extent, innovative services are being enabled and/or frustrated by the application of existing copyright law to software in everyday products

In recent years, device makers and service providers have developed no shortage of new and innovative offerings. From media streaming services like Netflix and Spotify to wearable technology like Fitbit and the Apple Watch, developers have found new ways to deliver goods and services to meet the demands of consumers.

But none of these goods and services depend on copyright law continuing down the dangerous path outlined above. Although device makers are eager to limit competition, control consumer behavior, restrict secondary markets, and extract higher prices for services, consumables, and replacement parts, there is no reason copyright law should indulge those whims. First, it is far from clear that the development and introduction of these new products depends on this degree of downstream control. And to the extent such control is necessary, it is available through other means that do not conscript the Copyright Act into the effort to gain control over the market for devices and related services. That is not the job of copyright law, nor should it be. If device makers want to exert that control, they can turn to other bodies of law including patents and contracts. Even within the copyright framework, if it is essential to control post-sale use of a product, device makers and service providers have a clear and workable option—don't sell your product to consumers. Products can be offered as rentals, leases, or subscriptions—transactions that consumers understand do not confer the rights associates with ownership. But when a consumer buys a car, a phone, or a pacemaker, they expect to own it. And that expectation is not limited to the physical components; it reasonably includes the software that is equally, if not more, responsible for the device's characteristics, features, and performance.

4. Whether, and to what extent, legitimate interests or business models for copyright owners and users could be undermined or improved by changes to the copyright law in this area

The current state of the law surrounding software-enabled devices poses a significant risk to a number of legitimate interests and business models. It threatens secondary markets and intermediaries like eBay that facilitate resale. It threatens portions of the sharing economy, including sites like Neighborgoods, that enable the lending of tools and other equipment. It threatens independent service and repair shops across the electronics, telecommunications, automative, and other sectors. It threatens the makers of low cost add-on products and compatible consumables, including amateur user-innovators.

Ironically, the uncertainty surrounding exactly what rights consumers acquire in the devices they buy could dissuade them making such purchases in the future. This assumes that a sufficient number of consumers are aware of the limited set of rights they acquire when it

[34] See Sega, 977 F.2d at 1522; Sony Computer Entertainment, Inc. v. Connectix, Corp., 203 F.3d 596, 602-03 (9th Cir. 2000); Lexmark Int'l, Inc. v. Static Control Components, Inc., 387 F.3d 522, 535-36 (6th Cir. 2004); Computer Associates International, Inc. v. Altai, Inc., 982 F.2d 693, 709-10 (2d Cir. 1992).

comes to “licensed” devices. And while there have been notable consumer backlashes to some instances of over-reaching rights holders, there is good reason to suspect that the public at large remains in the dark about the assertions rights holders make about their ongoing right to control devices after their sale.

Given the troubling consequences of allowing copyright holders to assert control over software embedded in devices sold to consumers, in conflict with the text and purpose of § 117, rights holders should be expected to demonstrate a concrete and significant risk of meaningful infringement of their software code. But they can’t. No one buys a John Deere tractor to extract its software code. No one buys a pacemaker to share its operating system via BitTorrent. Assertions of copyright in this space are not driven by a desire to prevent or discourage infringement; they are motivated by a desire to control how consumers use utilitarian devices, to limit competition and increase prices by curtailing secondary markets, and to leverage the limited copyright monopoly into the service and repair markets. Not only is that far from a compelling justification for copyright exclusivity, it is the definition of copyright misuse.

5. Key issues in how the copyright law intersects with other areas of law in establishing how products that rely on software to function can be lawfully used

In the context of software-enabled devices, copyright law intersects with at least four other bodies of law.

First, to the extent copyright law interferes with the ability of device owners to use, transfer, and repair the products they buy, the balance between the statutory rights of copyright holders and the personal property rights of consumers are implicated. Copyright law’s exhaustion rules are intended to achieve a balance between those conflicting sets of interests.[35] But to the extent copyright law fails to acknowledge consumers as the owners of not only the physical devices they buy, but also the software copies responsible for the functionality, the copyright system is responsible for a massive erosion of personal property rights.

Second, since license terms that purport to prevent a transfer of ownership are often characterized by licensors—and interpreted by courts—as contracts, the intersection of copyright and contract law is a necessary consideration.[36] Contract law has undergone a transformation in recent decades that replaced the traditional paradigm of negotiated terms, a meeting of the minds, and explicit acceptance with a contemporary view that permits contract formation by notice alone, and constructive notice at that. Under those conditions, contracts no longer create purely in personam rights. Instead, they effectively attach restrictions that run with a piece of personal property; they create servitudes on chattels, a property restriction that faced hostility at common law historically.[37] These so-called agreements seek to enlist copyright law in achieving an outcome that courts have rejected for centuries. Putting aside issues of formation, this suggests that license agreements imposing restrictions on the use of a device

[35] See Aaron Perzanowski & Jason Schultz, Reconciling Intellectual and Personal Property, 90 Notre Dame L. Rev. 1211 (2015).

[36] There are good reasons, however, to resist the notion that licenses are at their core contractual in nature. See Christopher M. Newman, A License Is Not A "Contract Not to Sue": Disentangling Property and Contract in the Law of Copyright Licenses, 98 Iowa L. Rev. 1101 (2013).

[37] See Molly Shaffer Van Houweling, The New Servitudes, 96 Geo. L.J. 885 (2008).

or its embedded software should be enforced not with copyright remedies, but with contractual ones—subject to internal contract law limitations and copyright preemption.

Third, false advertising is a relevant area of law that is typically overlooked in this discussion. Recent research suggests that when consumers buy digital goods, they often expect to acquire the same sorts of rights to use and transfer those goods that they acquire when they purchase analog products.[38] By characterizing transactions as sales or purchases when, in fact, the fine print imposes significant and unexpected limitations, device makers are misleading consumers about the fundamental nature of the transaction. At the very least, the law should ensure that consumers are able to make informed choices.

Finally, software-enabled devices present an increased potential for conflict with patent law. Historically, copyright law has made great effort to avoid extending protection to utilitarian devices. This is evident not only in § 102(b), but also on the useful article limitation on protection for pictorial, graphic, and sculptural works.[39] But if rights holders can exercise the kinds of control over devices described above by virtue of their copyright interests in the software that operates these devices, copyright has effectively embraced protection of utilitarian articles as such. This raises the possibility of overlapping copyright and patent protection for the functional features of the same device, not to mention the potential for copyright protection for devices that fail to meet the more stringent standards of possibility—protection that persists decades longer than the comparatively short patent duration.

Recommendations

Because the problems described above result primarily from misinterpretation of the terms of §§ 101, 109, and 117, legislative intervention is not strictly necessary. But courts clearly require some guidance regarding the fixation's durational requirement in order to put to rest the legacy of *MAI v. Peak*. Similarly, the courts would benefit from some clarification regarding the impact of license terms on the question of whether a "sale or other transfer of ownership" has occurred for purposes of determining who counts as an "owner" of a copy under §§ 109 and 117. The Copyright Office is well positioned to offer that interpretive guidance. In doing so, it should follow the lead of the Second Circuit in *Cartoon Network* and emphasize the distinct statutory durational requirement ignored in *MAI v. Peak*. Likewise, It should encourage an approach to resolving the license/sale question that looks to the economic reality of a transaction rather than the self-serving language of license terms drafted by copyright holders, as the Ninth Circuit did in *Augusto* and Second Circuit did in *Krause v. Titleserv*.[40] These recommendations apply with equal force to software and non-software disputes.

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[38] See Aaron Perzanowski & Jason Schultz, The End of Ownership (MIT Press, forthcoming 2016) (describing survey data that reveal a substantial percentage of consumers believe that they acquire the rights to resell, lend, and give away digital books, music, and movies after clicking the ubiquitous "Buy Now" button).

[39] 17 USC § 101(defining a "useful article" as one "having an intrinsic utilitarian function that is not merely to portray the appearance of the article or to convey information" and providing that "the design of a useful article ... shall be considered a pictorial, graphic, or sculptural work only if, and only to the extent that, such design incorporates pictorial, graphic, or sculptural features that can be identified separately from, and are capable of existing independently of, the utilitarian aspects of the article").

[40] 402 F.3d 119 (2d Cir. 2005)

If legislation is necessary, promising models have been proposed in Congress and enacted in other jurisdictions. The You Own Devices Act (YODA)[41] addresses some of the most troublesome implications of current copyright law as applied to software-enabled devices. It would entitle the owner of a device to transfer an authorized copy of a computer program that enables operation of that device when the owner sells, leases, or otherwise transfers the device. That right cannot be waived. YODA would greatly improve the current state of uncertainty regarding transfers of devices, but it could go further by giving owners the right to modify their software, as currently permitted under § 117.

A statutory definition of a "transfer of copy ownership" or an "owner" for exhaustion purposes could also offer much needed clarification. To the extent those terms are defined, they should recognize that a transaction characterized by a one-time payment in exchange for perpetual possession of or access to a copy should be considered a sale regardless of the terms of a license.

When it comes to the RAM copy doctrine, existing legislation in Canada and Israel provide useful starting points. In 2012, Canada added § 30.71 of the Copyright Act which provides that it is "not an infringement of copyright to make a reproduction of a work or other subject-matter if (a) the reproduction forms an essential part of a technological process; (b) the reproduction's only purpose is to facilitate a use that is not an infringement of copyright; and (c) the reproduction exists only for the duration of the technological process." Similarly, Copyright Act,. § 26 of the Israeli Copyright Act provides: "The transient copying, including incidental copying, of a work, is permitted if such is an integral part of a technological process whose only purpose is to enable transmission of a work as between two parties, through a communications network, by an intermediary entity, or to enable any other lawful use of the work, provided the said copy does not have significant economic value in itself." By incorporating a similar provision in the Copyright Act, much of the harm of the RAM copy doctrine could be avoided.

The Request for Public Comment asks "whether copyright law should distinguish between software embedded in 'everyday products' and other types of software." Since the problems highlighted here have their origins in cases dealing with traditional computer software, it would be unwise to distinguish software embedded in everyday products from software installed on traditional computers. While the concerns are more obvious and startling when we talk about software-enabled devices, those same worries exist for software installed on desktops and laptops. In both cases, the takeaway is the same—when a product is sold to consumer, she should be entitled to treat that product as her property, as defined by §§ 109 and 117. If the rights of public are contingent on the kindness of licensors, in the near future copyright law will regulate nearly every device with which we interact.

Respectfully,

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[41] H.R. 862, 114th Cong. (2015)

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